

UNIT-1

- Types of Data Structures (With Figure).
- Revision of Pointers, DMA, Structures and Unions.
- Advanced Pointer Concepts - Types.
- Memory Address Calculation - Row Major Order, Column Major Order.
- Algorithm Analysis Concepts - RAM Model.
- Searching - Linear and Binary Search
- Sorting - Bubble and Selection Sort.

UNIT-2

- Stacks - Definition, Operations - push, pop, peek.
- Infix to Postfix and Postfix Evaluation (Algorithm + Questions).
- Applications of Stacks.
- Recursion - Types, Recursion Vs Iteration Table, Examples - GCD, Binomial Coefficient, Towers of Hanoi
- Queues - Definition, Types.
- Operations - Simple Queue.

UNIT-3

- Linked List - Definition and Representation.
- ADTs of Singly Linked List (SLL).
- Doubly Linked List (DLL) and its operations (ADTs) .
- Circular Linked List (CLL).

UNIT-4

- Graphs - Definition and Types.
- Graph Representation - Adjacency Matrix & Adjacency List.
- Operations on Adjacency Matrix and List - Insert / Delete Edge, Insert / Delete Vertex, Edge Lookup
- BFS and DFS (Algorithms)
- Trees - Definition and Types.
- Binary Trees - Definition and variants.
- Operations on Binary Trees
- LCRS Representation.
- 3 types of traversal - preorder, inorder and postorder.
- Binary Search Tree and operations.

RED - MOST IMPORTANT TOPICS.